# Towards IoT-Driven Predictive Business Process Analytics

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***Abstract*— Predictive business process monitoring is concerned with predicting the process-related Key Performance Indicators (KPIs) and forecasting the future behavior of the process in realtime. Despite the amount of work contributed by researches to this field of research, the performance of existing solutions is not desirable for practical settings. Indeed, these approaches are typically context-unaware and lack generality. However, in real-life use cases, business processes are not isolated from the surrounding working environment, and thus they are influenced by many contextual events, such as events generated by IoT devices. To the best of our knowledge, there is no comprehensive study addressing the integration of contextual events with the process prediction. This paper proposes a holistic context-aware methodology for predictive process monitoring by incorporating IoT data. Moreover, we present a systematic method to integrate the contextual events in the runtime process using Business Process Management System (BPMS) capabilities. We also introduce a predictive model based on Deep Neural Networks (DNN) to forecast the next activity. Finally, we evaluate our solution using a case study in the aviation industry.**

*Index Terms*—**Business Process Monitoring, Predictive Analysis, Context-Aware, IoT, Process Mining.**

## I. INTRODUCTION

Today, business processes create the essence of any business and hire all business entities, assets, and efforts to achieve the enterprise goals. Continuous process monitoring is a vital part of any business that analyzes the performance of process execution. Novel approaches in process monitoring are known as Predictive Business Process Monitoring. Predictive process monitoring (PPM) is a subset of process mining techniques. It is concerned with predicting how the running cases will complete. The previous monitoring methods are reactive. It means they reveal problems only after they have occurred [1]. However, PPM is a proactive approach that uses Machine Learning (ML) algorithms. Note that business processes are not isolated from the environment, and their performance have been influenced by the contextual events such as events that come from the Internet of Things (IoT). Since these kinds of events impact on process execution and change the process behavior, they should be taken into account for process analysis. To the best of our knowledge, there are hardly any approaches to consider context data in business process monitoring.

Context-aware PPM can have many applications in the healthcare domain [2]. Healthcare systems regularly collect process data of diagnosing and treating diseases. IoT events are another part of the data collected in the healthcare domain that continuously record the healthcare process context (such as heart rate, blood pressure, etc.). The treatment process is influenced by IoT devices in a hospital, and the decision takes a stand on IoT events monitoring. Since IoT events do not record in the process event logs directly, they do not involve in the process prediction. Easy to perceive that linking IoT events to the treatment process steps can provide insightful predictions.

A business process is a set of activities and decision points that are done by actors such as people and machines. Information systems record every traces of the process execution. Every process footprint called a process instance or a case, and historical cases extract as event logs. PPM takes a set of historical completed cases as input data and then use ML algorithms to build prediction models. At runtime, trained prediction models apply to event streams for ongoing case prediction [3].

Process events emitted when an activity is performed [4]. Process data can be divided into two groups of intrinsic events and contextual events. Intrinsic events are generated by performing process steps and recorded as process event logs. Contextual events are collected from the third-party process data resources such as IoT devices and other information systems, and linked to the process, indirectly [5]. To the best of our knowledge, the previous works have focused on the exploitation of events that are recorded directly in event logs. Only a few works consider contextual events in the

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process prediction approaches [5]–[13]. There are not specific integration solutions for using contextual events in the PPM [6]–[10]. To acquire context data collected with modern data sources (such as IoT devices) in PPM, first, it is necessary to collect and integrate the contextual events with the process data. In [5], [12] a limited attention has been paid to the study of context data integration operations in the process.

Events can not consider in isolation, and a correlation between them should be developed [14]. Contextual events should be incorporated into the business process. For example, IoT events can be linked to the intrinsic events using temporal information (as shown in Fig.1(A)) or using expert knowledge [5]. Our proposed approach integrates IoT events into process logic and exploit these events in the process execution (as shown in Fig.1(B)). IoT events after collection from process context are invoked in the running cases and involved in the process execution. We use the contextual event in the process decision point to incorporate it into the process execution logic. As a result, context changes the process behavior and its path, and then directly recorded in the event logs.

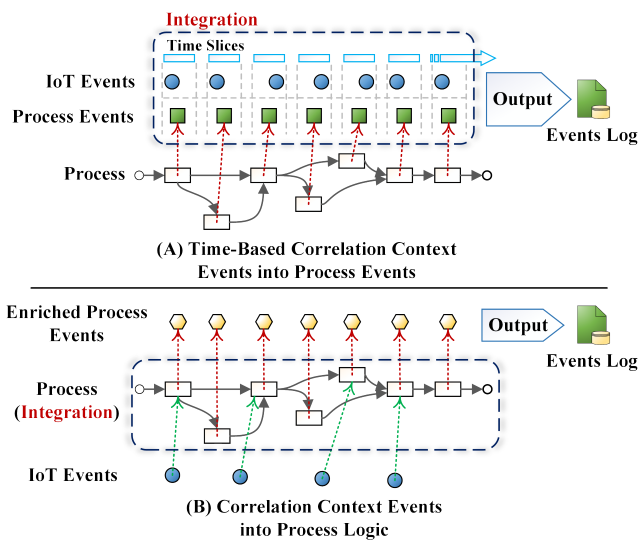


Fig. 1: The Comparison of Context Data Integration Methods in the Process Data.

The rest of this paper is organized as follows. Section 2 discusses the related works. Section 3 introduces the proposed approach and building blocks to support the idea behind it. Section 4 presents the experimental results of simulation and implementation of the proposed approach. Finally, Section 5 concludes this paper and introduces future works.

## II. RELATED WORK

In this study, inspired by [14], the previous works in business process prediction field are categorized in four groups according to the output of the prediction. In addition, we have added a new group for the rest of related works that cannot be placed in the previous categories. The new group of studies focuses on different aspects of PPM. Table I summarized the existing solutions and presented in the following five categories: (1) Time-Based Predictions, (2) Process Output Prediction, (3) Process Path Predictions, (4) Process Risk Predictions, (5) Other Predictions and Works. The columns of the table include the reference number of the study, the prediction output, the algorithms used for the prediction, the implementation environment, the industry or business domain that the approach belongs to, the input data type, being a context-aware approach, incorporating IoT data in the prediction approach, and the availability of the dataset.

As illustrated in the table, most of the existing works belong to Time-Based Prediction category [6]–[9], [15]–[22]. In this category, prediction of the remaining time is the most popular goal [6]–[8], [16], [18]–[20], [39]. The decision tree is the most popular algorithm for building predictive models [13], [14], [16], [27], [31]. It seems that the new trend in recent studies is using the ANN in PPM [5], [39]. Those studies have used ANNs to improve prediction accuracy. In most previous works, the proposed approaches implemented as a plug-in on top of the ProM[1] framework [5], [7], [9], [11], [13]–[15], [17], [18], [20], [21], [24], [27], [28], [32], [37], [40]. The ML algorithms developed in the Weka[2] toolkit [7], [12], [13], [18], [21]. In [21], [37], [40], CPN[3] tools and in [23], Java have used for the process simulation and generating synthetic event logs. Most of the studies have used several real-world datasets on transportation and the financial domain. Business Process Intelligence Challenge[4] (BPIC) datasets that publicly released have used in most of the works [5], [12], [13], [29], [32], [39]. Note that most of the datasets are not available. Differences in prediction goals, datasets, and also limited access to the datasets make the comparison of the studies quite difficult.

Previous works such as [11], [41], [42] have confirmed the importance of the contextual events in processes and their relevance to the process activities as well as events characteristics. Those data have exploited in some of the previous works [5]–[13]. However, there is no specific proposed approach or architecture so far that supports the required steps of collect, integrate, preprocessing, and processing contextual event data into a PPM framework. In fact, no attention has been given to collecting and integrating contextual events to the process entities (i.e process events, activities, decisions, and traces) [12]. The studies [12] and [5] have highlighted the lack of published open datasets containing IoT events. Recently in [5], the authors have proposed to extract the contextual events from the modern resources such as Business Intelligence processes and IoT events and then use them in the process analysis.

In [6], context information have been used to separate performance-relevant variants presented by regression models. The paper [7] considers different data for each container and physical characteristics as context data. In [8], the authors proposed a predictive-clustering approach to identify different scenarios related to the process context, and equip them with

[1]http://www.promtools.org/
[2]https://www.cs.waikato.ac.nz/
[3]http://cpntools.org/
[4]https://www.win.tue.nl/bpi/doku.php?id=2018:start&redirect=1

TABLE I: The Existing Studies in Business Process Prediction.

| | Prediction | Algoritems | Implementation | Domain | Input Data | | Context-Aware | IoT | Data Access |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | Real-World Data | Artificial Data | | | |
| | | | | Time-Based Predictions | | | | | |
| **[15]** | Completion Time and Next Activity Prediction, | C4.5, M5 and Sequential Pattern Mining | ProM | Communication Company | ✓ | - | - | - | - |
| **[16]** | Next Step, Remaining Time and Possible Deadline Violations | Decision Trees and Decision Mining | CoCaMa | - | - | - | - | - | - |
| **[17]** | Activity Durations and Decision Probabilities | Time Series | ProM and R | Call Center | ✓ | ✓ | - | - | - |
| **[18]** | Remaining Time | Naive Bayes Classifiers and Support Vector Regression | ProM and Weka | Road-Traffic Fines | ✓ | - | - | - | - |
| **[19]** | Remaining Time | Multivariate Regression and Hidden Markov Model | R | - | - | ✓ | - | - | - |
| **[20]** | Remaining Time | Stochastic Petri Nets | ProM | Logistics | ✓ | ✓ | - | - | - |
| **[6]** | Remaining Time | Pattern Mining, Non-Parametric Regression and Predictive Clustering | - | Transshipment | ✓ | - | ✓ | - | - |
| **[21]** | Average Process Durations | workflow Simulation by Petri nets | ProM and Weka | Credit Card Application | - | ✓ | - | - | - |
| **[22]** | Delay Prediction | Queue Mining | R | Telecommunications and Financial Sectors | ✓ | - | - | - | - |
| **[7]** | Remaining Processing Time and Steps | Predictive Clustering, Pattern Mining and Predictive Clustering Tree | ProM and Weka | Transshipment | ✓ | - | ✓ | - | - |
| **[8]** | Remaining Time | Predictive Clustering Tree | - | Transshipment | ✓ | - | ✓ | - | - |
| **[9]** | Waiting Times, Throughput Times, and Utilization Rates | ANOVA | ProM | Financial Institution | ✓ | - | ✓ | - | ✓ |
| | | | | Process Output Prediction | | | | | |
| **[23]** | Outcomes at Decision Points | Ant-Colony Optimization | Java | Insurance | - | ✓ | - | - | - |
| **[24]** | Future of a Running Case (Properties of Events) | Predictive Clustering Tree | ProM | Journal Reviewing, Repair and Insurance | ✓ | - | - | - | ✓ |
| **[25]** | Outcome Cases | Text Mining with Sequence Classification | Python | Debt Recovery and Lead-to-Contract | ✓ | - | - | - | - |
| **[10]** | Performance Measure (Remaining Processing Time and Steps) | Non-Parametric Regression and EM Algorithm | Java | Transshipment | ✓ | - | ✓ | - | - |
| **[26]** | Aggregate Performance Outcomes | Time-Series Models | - | Transshipment | ✓ | - | - | - | - |
| **[27]** | Goal Achievement | Decision Tree | ProM and Python | Repair | ✓ | - | - | - | ✓ |
| **[28]** | Process-Related Costs | - | ProM | Repair | - | ✓ | - | - | - |
| **[29]** | Next Event | EM Algorithm and Probabilistic Model | - | Financial Institution | ✓ | - | - | - | ✓ |
| **[30]** | Potential Problems | Support Vector Machines | Python | Logistics | ✓ | - | - | - | - |
| | | | | Process Path Predictions | | | | | |
| **[31]** | Execution Path | Decision Trees | SPSS Modeler | Customer Marketing | - | ✓ | - | - | - |
| **[32]** | Sequences of Events | Probabilistic Finite Automaton and Maximum a Posteriori | ProM | Financial Institute | ✓ | - | - | - | ✓ |
| **[33]** | Next Process Step | Categorical-Sequential Clustering and Markov Models | MATLAB | Telecommunications | ✓ | - | - | - | - |
| **[34]** | Next Event | Probabilistic Finite Automaton and EM Algorithm | Java | Incident and Problem Management | ✓ | - | - | - | - |
| **[35]** | Next Step | Markov Models and Sequential K Nearest Neighbors | MATLAB | Telecommunication | ✓ | - | - | - | - |
| | | | | Process Risk Predictions | | | | | |
| **[36]** | Risk-Informed Decision | C4.5 Algorithm | YAWL BPM system | Insurance | ✓ | - | - | - | - |
| **[37]** | Overall Process Risk | Non-Parametric Regression and Time Series | Prom and R | Financial Institute | ✓ | ✓ | - | - | ✓ |
| **[38]** | Propagates Risk | Similarity-Weighted Process Instance Graph (PING) and Risk Propagation Algorithm | Workflow Management System Camunda Extension | Financial Institute | ✓ | - | - | - | - |
| | | | | Other Predictions and Works | | | | | |
| **[11]** | Characteristics of Events | C4.5 Algorithm | ProM | Insurance | ✓ | - | ✓ | - | ✓ |
| **[14]** | Characteristics of Events | Decision Tree and Regression Tree | ProM | Insurance | ✓ | - | - | - | ✓ |
| **[5]** | Event-Based Failure | Neural Networks | ProM | Financial Institute | ✓ | ✓ | ✓ | ✓ | ✓ |
| **[39]** | Next Task, Remaining Time | Long Short-Term Memory (LSTM) Neural Networks | Python | Help Desk and Financial Institute | ✓ | - | - | - | ✓ |
| **[40]** | Completion Time | Annotated Transition System | ProM | Transshipment | ✓ | ✓ | - | - | - |
| **[12]** | Violation of Aggregate Performance | Regression-Based Time-Series Predictor (TSPM) | Java and Weka | Transshipment and Problem Management | ✓ | ✓ | ✓ | - | ✓ |
| **[13]** | Estimating The Probability Outcome | Decision Trees and Random Forests | Prom and Weka | Hospital | ✓ | - | ✓ | - | ✓ |

the process performance model. The proposed approach in [9] analyzes key process performance indicators by considering the process context. This work examines the execution interval of an activity based on its context, and calculates the resources involved in an activity. Then, it determines the influence of the involved resources on the execution time. This process has been done by applying context functions to the activity. The method presented in [10] also uses different data which are related to the containers as context data for utilizing in the process analysis.

In [13], a framework has been indicated and evaluated by the patient's treatment process in a Dutch hospital (BPIC 2011). It uses some process features such as patient age as the context data. The paper [11] shows how process characteristics can be derived and linked to different events, including processes and context data. The authors of [14] update the public framework presented in the previous work and use context data as a resource to insert new features to the process and enrich the event logs in order to have a better analytic. In [5], the authors proposed a method for event-based failure prediction in the distributed business processes. This approach allows making use of traditional process management systems, event-based systems, as well as context data to improve the prediction output. However, the details of using contextual data in the proposed architecture have not been clearly determined. The method presented in [12] was proposed to predict the performance of the process using the running processes. The architecture proposed in this paper is designed based on the data collected from the process and the environment. This method exploited the process and context data in specific time windows for prediction process.

Altogether, no serious attempt has been made to support transferring IoT events to the process analysis. Only a few methods have been presented to integrate the context data with the process.

## III. PROPOSED APPROACH

This section presents our proposed approach for context-aware PPM. The basic idea of our approach is using IoT events as a process context. As shown in Fig.2, the steps of the proposed approach are as follows: (1) Data Collection, (2) Data Integration, (3) Data Preprocessing, (4) Data Processing, and (5) Presentation.

- *Data Collection:* In this step, the IoT events are collected and become ready for integration with the process data. Note that some initial data adaptations may occur in this step.
- *Data Integration:* In this step, to enrich process data, contextual events integrate with them in the process logic at runtime. In fact, IoT events affect underling principles, assumptions, and rules of the business process.
- *Data Preprocessing:* In this step, process data are adapted to the requirements of data processing to generate predictions. For example, data may need to convert to XES format.
- *Data Processing:* In this step, enriched process data is used to predict business processes. Note that process event logs are used for training the prediction models (offline) and at runtime, the enriched event streams are used for real-time prediction.
- *Presentation:* In the final step, the prediction of running cases are shown in the graphical user interface (GUI). They can be used in a recommender system, or a notification system. Additionally, predictions can be reused in the process engine to the behavior of the running cases.

Fig.3 shows the building blocks of the proposed context-aware PPM approach. The structure of this figure consists of six layers that support the required operations within the five steps above. The idea of these layers comes from Lambda Architecture [43]. Contextual events are collected from the process environment by IoT devices in the *IoT layer*. The *Stream Processing Layer* acquires real-time event streams, performs some data transformations, and transfers the data to the *Data Integration Layer*. In the *Data Integration Layer*, the process engine fetches the contextual events and integrates them into the process execution. The *Batch Layer* is responsible for storing and serving persistent data that are useful in training prediction models. In the *Predictive Process Monitoring Layer* the prediction models are built. In this layer the enriched process events are used to predict the business processes. The results of the predictive analysis are shown through the *Presentation Layer*.

### *A. Internet of Things Layer (Process Context)*

IoT devices have a significant role in collecting valuable data from the environment. Data gathered from the devices are transmitted to the cloud for analytics. In fact, environmental changes can be delivered into the business process by incorporating the IoT data. High-velocity, high-volume, and high-variety of IoT data are the properties of the contextual data that demand innovating forms of data processing.

### *B. Stream Processing Layer*

The main challenges of hiring IoT events to predict business processes are dealing with the big data streams and preparing IoT data to integrate with the process data. To overcome these challenges, we considered a building block for the Stream Processing. This layer not only facilitates the online processing of IoT data, but also it handles data adaptation for integration requirements. In addition, the Stream Processing layer can be used to manage dynamic changes in the number of data producers (IoT devices) and even data consumers (running cases). This layer contains a Message Broker for routing event streams and an Event Processing Unit (EPU) for data adaptation and some data preprocessing before integration.

### *C. Data Integration Layer (BPMS)*

For context-aware PPM using IoT evens, we need to incorporate the process characteristics with IoT events. In our

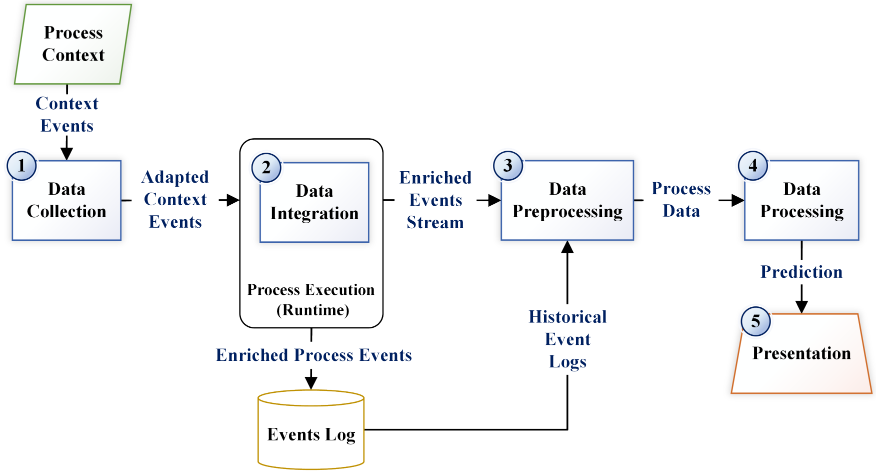


Fig. 2: The Proposed Context-aware PPM Approach.

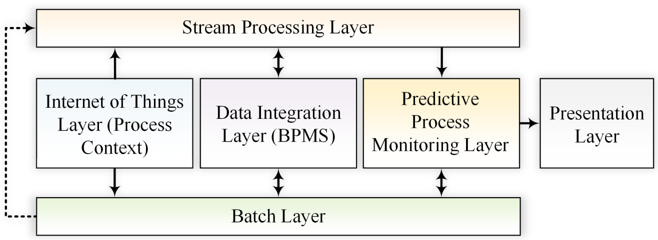


Fig. 3: The Building Blocks of the Proposed Approach.

approach, the enriched process event logs have been created by the process engine of BPMS. The process engine acts as an integration point that incorporates the contextual events with process characteristics. According to [44] and [45], the key to a precise process analysis is defining a meaningful relationship between the contextual events and the process. This leads to more valuable and insightful prediction.

We use contextual events in the process decision point to control the process execution path, and incorporate contextual changes with process rules and exaction logic at the runtime. The process engine receives IoT events through the Service Tasks (BPMN[5] 2.0), and then utilizes them to affect process behavior. Actually, our approach allows making use of BPMS and their capabilities to integrate IoT events with the process data. Most of the modern BPMSs are capable of invoking Web Services (RESTful or SOAP) and connecting to data sources. Subsequently, we use this feature to transmit the contextual events to the process level and integrate them into the process execution. Hence, IoT events directly participate in the process analysis. Note that sometimes we need a domain expert to define the relations, and process redesign to integrate the contextual events into the process.

[5]http://www.bpmn.org/

### D. Batch Layer

The Batch layer represents all persistent data in the system. The data needs to store for further processing, the process data (enriched with contextual events), the process models, and prediction models are kept in this layer.

### E. Predictive Process Monitoring Layer

The PPM layer makes use of event streams to predict incomplete cases. To predict business processes, this layer uses both the trained prediction models and event streams of running cases. The former are received from the Batch layer, and latter from the Integration layer.

ML algorithms regularly receive enriched event logs and update the prediction models. The Stream Processing layer can use to send complete traces on time, and update the prediction models as each process instance is terminated.

At last, a set of predictions and process analysis are presented as the output. The Presentation layer contains groups of graphical reports, management alerts or APIs.

## IV. EXPERIMENTAL RESULTS

We illustrate the applicability of our proposed approach using a synthetic event logs inspired by the case study [46]. For the experiment, event logs of the aircraft take-off process have been used. As shown in Fig.4, we have modeled the simplified take-off process, in terms of a Business Process Model and Notation (BPMN) diagram. This process flow contains seven activities and a decision point. For instance, in the general scenario, the pilot fills the flight plan, and a case begins. Then, the flight plan is delivered to the Air Traffic Control (ATC) unit, which includes process information (such as the pilot's information, airplane information, and flight information). The ATC analyzes the flight plan for each case. They review flight plans based on the rules and regulations and other conditions like weather. The weather conditions are one of the significant factors associated with the take-off process (as shown in the take-off process domain model [46]). Nowadays, IoT devices at the airport continuously collect the changes in weather.

However this kind of data has involved in the flight process, indirectly. In this paper, a service task has been added to the take-off process model to retrieve IoT events. As a result, the process contextual events are captured during the runtime of a process step. Those events have been used in the process decision point to refine the process behavior. Indeed, we defined a set of rules for the process decision point that directly incorporates the weather change into the process execution logic and path. The flight plan will be accepted or rejected based on the exclusive gateway. This loop will repeat until the predefined rules based on flight information and weather conditions are met. Finally, the aircraft can take-off after it is granted permission to flight.

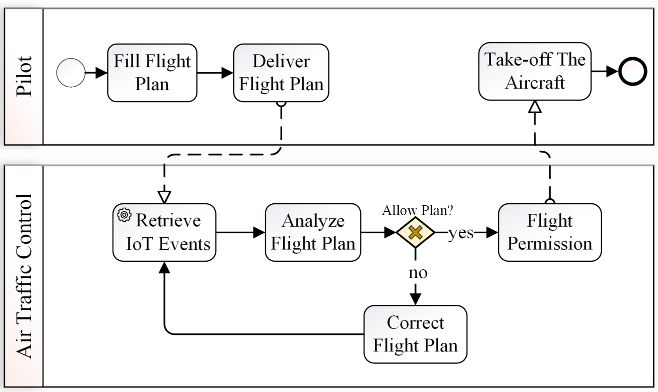


Fig. 4: The Simplified Aircraft Take-off Process Model.

The aircraft take-off scenario and process rules have been implemented in Java, and the synthetic event logs have obtained via simulation. Moreover, IoT events have been generated for three weather sensors. Note that we use Apache Kafka[6] to implement Stream Processing Layer, which provides a publish and subscribe platform for IoT event streams. Our goal is to predict the next activity built on the process scenario and the process model at the decision point.

The analysis has been performed using a synthetic take-off event logs containing 1000 cases and 7887 events. The event logs contain the process intrinsic information (*Event ID, Case ID, Activity, Timestamp, Flight Number, Pilot Grade,* and *Plane Type*), as well as the contextual events (*Temperature, Wind Speed,* and *Humidity*). We used 80% of the traces as the training set and the remaining as the test set. The construction of the prediction model relies on the implementations of the ANN. We use Python's Scikit-Learn[7] library to create our ANN that performs classification. According to the recorded event logs, feature vectors contain pilot grade, aircraft type, and weather information have been used as inputs for the classification model, and *Flight Permission* and *Correct Flight Plan* labeled as the response variable for the supervised learning. The ANN used for classification had five inputs, and one output to predict the response variable, and two hidden layers with five and three neurons. The classifiers are trained with prefixes of historical traces, and used to predict the next activity at the process decision point.

[6]https://kafka.apache.org/

[7]https://scikit-learn.org/

TABLE II: The Evaluation Metrics for the Classifier (Flight Permission labeled 1 and Correct Flight Plan labeled 0).

| Label | Precision | Recall | F1-score | Count |
|---|---|---|---|---|
| 0 | 0.89 | 0.92 | 0.91 | 507 |
| 1 | 0.95 | 0.93 | 0.94 | 796 |
| Accuracy | | | **0.92** | 1303 |

The confusion matrix has been used to evaluate the quality of the output of the classifiers on the training dataset. As shown in Table II, we used three metrics of Precision, Recall, and F1-score to evaluate the prediction models. The accuracy of the model is 92% on the training set, and 91% on the test runs. For prediction at runtime, the constructed classifiers can be applied to the extracted feature vectors from a prefix of an ongoing trace.

## V. CONCLUSION

Predictive business process monitoring attempts to predict the future of the running cases. Business processes have been influenced by the contextual events such as events generated by the IoT devices. Context-aware PPM has many applications in different businesses and industries. However, only a limited amount of approaches covered all the steps required to incorporate the contextual events into the process prediction. In this paper, we mainly focused on highlighting primary steps for using IoT events in process prediction. Additionally, we propose to integrate the contextual events with the runtime process, for a powerful correlation, and then specified the characteristics of building blocks needed for performing the proposed approach. The approach has been evaluated using a case study within Air Traffic Control. The synthetic event logs obtained from the aircraft take-off scenario. Then enriched event logs included IoT events exploit for context-aware prediction. In future work, we intend to make a more extensive architecture for using modern data resources in PPM.

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